$$f(rax, cl_0) = rax_0 \cdot 2^{-(cl_0 \cdot 2 + 1)}$$

$$!!\ cl_0 < 127\ !!$$

```
mov     ch, cl
inc     ch
shr     ch, 1
shr     cl, 1
shr     rax, cl
xchg    ch, cl
shr     rax, cl
```

$ch := cl_0$

$ch = ch + 1 = cl_0 + 1$

$ch >> 1$ ie $ch = (cl_0 + 1) >> 1$

$cl >> 1$ ie $cl = cl_0 >> 1$

$rax >> cl$ ie $rax = rax >> (cl_0 >> 1)$

ie $ch = cl_0 >> 1$

$cl = (cl_0 + 1) >> 1$

$rax = rax >> cl$

$= (rax >> (cl_0 >> 1)) >> cl$

$>>$ order of ops

(i) P arrangement 1st

(ii) opp precedence EMDAS

(iii) $>>$: left-associative

ex) $x >> y >> z = (x >> y) >> z$

$$rax = (rax >> (cl_0 >> 1)) >> (cl_0 + 1) >> 1)$$

ie

$$x = (x >> \underbrace{(a >> 1)}_{\frac{a}{2^1}}) >> \underbrace{(a+1) >> 1)}_{\frac{a+1}{2^1}}$$

$$2^{-(a+\frac{1}{2})} \cdot x$$

$$\frac{x}{2^{\left(\frac{2a+1}{2}\right)}} = \frac{x}{2^{a+\frac{1}{2}}}$$

$$\frac{x}{2^{(a/2^1)}}$$

$$\frac{x}{2^{\left(\frac{a}{2}+\frac{a+1}{2}\right)}} = \frac{x}{2^{(a/2)}} \cdot \frac{1}{2^{\left(\frac{a+1}{2}\right)}}$$

[0x27]